
(a) Perception
Monocular video
Event detection
Keypoint detection
and needle segmentation
reposition end (t = 0.0 s)
entry (t = 4.6 s)
t = 0.7 s
t = 1.3 s
t = 2.5 s
t = 3.4 s
t = 4.6 s
needle mask
needle tip
grasp point
jaw tip / jaw base